Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/14/2005
PI Name:	Sutherland, Betsy M Ph.D.		
Project Title:	Complex Space Radiation-Induced DNA Damage Clusters In Human Cell Transformation: Mechanisms, Relationships And Mitigation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	11973-5000	**Congressional District:**	1
Comments:	PI Deceased as of October 2009.		
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/01/2003	**End Date:**	05/31/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NA		
COI Name (Institution):			
Grant/Contract No.:	T-467X		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Predicting and mitigating cancer risk from complex radiation fields in outer space is essential for manned exploration of Mars. This requires understanding the molecular mechanisms by which such radiation inflicts critical damages, responses of cells to radiation, and potentially cancer-inducing changes. Clustered DNA damages are composed of several close-spaced DNA alterations. Specific clusters have different biological impacts, depending on the ability of cells to repair them, and in their cancer-inducing potential. Their levels and types depend on the radiation type, and on factors such as the environment of the DNA. However, repair of clusters in cells exposed to simulated complex radiation fields, the effect of such radiation on induction of potentially oncogenic cellular alterations, and the optimum approaches for mitigating their effects are not known. The project focuses on four experimental series: 1. Determine if prior exposure of human cells to protons affects repair of HZE particle-induced double strand breaks, abasic-, oxidized purine- and oxidized pyrimidine clusters; 2. Test whether dual proton-HZE particle exposures affect neoplastic		

	transformation levels of human cells relative to exposure to each radiation species singly; 3. Determine whether levels of slowly-repaired or unrepaired clusters correlate with human cell neoplastic transformation frequencies; and 4. Test whether factors that reduce cluster induction or increase their repair also decrease levels of neoplastic transformation of human cells. Understanding and mitigating space radiation-induced damage to human cells requires understanding the mechanisms of DNA damage induction and cellular repair, and delineating potential relationships between DNA damage clusters and their cancer-incuding potential. These studies should identify approaches for mitigating potentially cancer inducing effects of space radiation exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the role of complex DNA damage clusters in killing and mutating cells can lead to better radiation therapy of cancers. Further, discovering approaches for decreasing DNA damages can minimize the harmful effects of radiation on normal cells.
Task Progress:	Transformation of human primary skin cells by different charged particles found in deep space has been measured as a function of radiation dose. These include protons (188 MeV/n, 233 MeV/n and 1 GeV/n), Si ions (600 MeV/n) and Fe ions (1 GeV/n). The dose response lines are linear, and replicate experimental data show excellent reproducibility. During space travel, cells of astronaut crews will be exposed to different kinds of charged particles. Protons are the most numerous among the particles found in space, but heavy ions like Fe are expected to have the most severe biological effect. Since cells of travelers will be exposed to many proton 'hits' and few Fe 'hits,' it is important to know the effects of exposure to both protons and Fe ions on cell transformation. Experiments were carried out at the NASA Space Radiation Laboratory (NSRL) in which human primary cells were exposed first to a low dose of protons followed by a single hit per cell of Fe ions. The time between the two radiation types varied from 2 minutes to 24 hr. The transforming effect of protons alone and of Fe alone was also determined. For cells exposed to protons followed within about 1 hr by a Fe 'hit,' much higher transformation frequencies were observed than would be expected from the simple additive effect of the two radiations. If the Fe exposure was given at longer intervals, the dual hit regime gave similar results to the simple additive effect of the two radiation exposures. These results will be repeated in the next NSRL run to determine their reproducibility. The molecular origins of these effects is being studied by measuring changes in cell cycle, in gene expression and in the induction and repair of complex DNA damages.
Bibliography Type:	Description: (Last Updated: 04/12/2011)
Abstracts for Journals and Proceedings	Sutherland BM. "Clustered DNA damages in mammalian cells." 8th International Workshop on Radiation Damage to DNA, Banff, Canada, May 2004. 8th International Workshop on Radiation Damage to DNA, Banff, Canada, May 2004. Abstract 37. , May-2004
Abstracts for Journals and Proceedings	Sutherland BM. "DNA Damage Clusters in Human Cells." Symposium in honor of Dr. Helen Evans, Mechanisms in Radiation Carcinogenesis, Cleveland OH, November 2004. Symposium in honor of Dr. Helen Evans, Mechanisms in Radiation Carcinogenesis, Cleveland OH, November 2004. Abstract 9. , Nov-2004
Articles in Peer-reviewed Journals	Sutherland BM, Cuomo NC, Bennett PV. "Induction of anchorage-independent growth in primary human cells exposed to protons or HZE ions separately or in dual exposures." Radiat Res. in press, March 2005. , Mar-2005
Presentation	Sutherland, B.M. "Complex space radiation-induced DNA damage clusters in human cell transformation" 3rd International Workshop on Space Radiation Research and 15th Annual NASA Space Radiation Health Investigators' Workshop, Port Jefferson NY. May-2004
Presentation	Sutherland, B.M. "Clustered DNA damages in mammalian cells" 8th International Workshop on Radiation Damage to DNA, Banff, Canada May-2004
Presentation	Sutherland, B.M. "Exploring the effects of space radiation on DNA in human cells" 2004 NASA Spaceflight and Life Sciences Training Program at the John F. Kennedy Space Center, FL Jun-2004
Presentation	Sutherland, B.M. "DNA Damage Clusters in Human Cells" Symposium in honor of Dr. Helen Evans, Mechanisms in Radiation Carcinogenesis, Cleveland, OH Nov-2004
Presentation	Sutherland, B.M., "Human cell neoplastic transformation by single ion beam- and mixed field irradiations" Bioastroanutics Investigators' Workshop, Galveston, TX Jan-2005